

Collection and identification of marine macroalgae in the rocky shores of coastal Karnataka, India

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ABSTRACT

Diversity of the seaweeds is threatened by the interplay of climate change and anthropogenic activities by expanding population and their demands. Apart from their economic values, they play an ecological role being the biggest carbon sinks. Thus, it is necessary to study and survey the diversity of marine algae in a timely manner. Moreover, the study of seaweeds from the Karnataka coast is very limited. Thus, the current study attempted to study the diversity of marine macroalgae along the coastal Karnataka. This study was conducted from April 2022 to December 2023, collection of marine macroalgae was done by a random pick method in rocky pools during low tides. Specimens were identified using morphological characters by referring flora, monograph and e-tools. Among the total 65 collections, 20 species belong to 6 genera of green algae, 27 species belong to 15 genera of red algae, and 18 species belong to 7 genera of brown algae were identified from 11 talukas belonging to 3 coastal districts, namely Uttara Kannada, Udupi, and Dakshina Kannada. The Uttara Kannada coast is rockier and has richer marine macroalgal diversity than Udupi and Dakshina Kannada. *Ulva*, *Chaetomorpha*, and *Sargassum* are dominant genera among the collection. Lesser diversity was recorded in the present study compared to earlier reports. Due to seasonal changes, strong wave action that causes seaweed drift during senescence, coastal erosion, sedimentation, and pollution factors, the macroalgal diversity along the Karnataka coast is declining.

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1. Introduction

Macroalgae are crucial organisms in marine ecosystems as primary producers in oceanic food webs. They are also called seaweeds due to their size, multicellular construction, and attachment to firm substrata, which are present throughout the world's oceans stabilizing coastal sediments (Al Ashwal et al., 2021; Cotas et al., 2023). The marine macroalgae-derived compounds have found applications in food, beverage, nutraceutical, cosmetic products, biomedical materials, pharmaceutical products, fertilizers, textiles, leather, medicine, biotechnology, and many other industries. Apart from their economic values, most importantly, they play an ecological role being the biggest carbon sinks (Qin, 2018; Saravanan et al., 2023). The lack of realization of the economic potential of these renewable marine resources, seaweeds have been underutilized in developing countries like India, though it has a long coastline of approximately 7500 km, constituting a 2.5 million km² economic zone and a 0.13 million km² shelf area. This could hold a vast diversity of habitats such as sandy beaches, rocky shores, deep tide pools, cliffs, estuarine mangrove vegetation, coral substrates, and artificial offshore structures, which support the growth of diverse seaweeds to thrive in such intertidal and subtidal waters. The highest number of seaweeds was reported from India, compared to the other countries that border the Indian Ocean (Mantri et al., 2020). Some of the pioneers, such as Iyengar (1927), Boergesen (1928), Umamaheswara Rao (1969), and Krishnamurthy (1980), contributed to increase the knowledge about the seaweed diversity of India. Ganesan et al. (2019) reviewed that Oza and Zaidi (2001) compiled a checklist of 844 species. Recently, many additions were made to the checklist of seaweeds of India, resulting in more than 900 species.

The diversity of these seaweeds in the current situation is threatened by the interplay of climate change and anthropogenic activities, including demands of an expanding population and unsustainable extraction of natural resources (De et al., 2024). Especially with the Karnataka coast in India, these macroalgal resources are increasingly threatened by urbanization, pollution, and tourism. As sensitive indicators of climate change, assessing their diversity is crucial for monitoring coastal ecosystem health (Harley et al., 2012). Thus, it is necessary to study the diversity of seaweeds in a timely manner. Moreover, the study of seaweeds from the Karnataka coast is very limited. A seaweed diversity and distribution study in Karnataka was established by Agadi (1985, 1986), in which forty-three species were reported. Continued by Untawale et al. (1989) who reported sixty-five benthic algae from the North Karnataka coastal regions. Later, Kaladharan et al. (2011) attempted to study coastal phytoplankton and marine algae, as well as flora, in which seventy-eight species of seaweeds were reported. Furthermore, Yadav and Palanisamy (2018) added *Martensia fragilis* from the Vannalli coast, and Yadav and Palanisamy (2020) added eleven marine macroalgae to the marine macroalgal flora as a result of exploration field studies during 2014-2017. In spite of many previous reports, the pictorial illustration of many seaweeds was missing, which is necessary for easy identification. Regular assessment of the diversity of macroalgae in Karnataka is not as vigorous as in other states like Tamil Nadu, Andhra Pradesh, and Gujarat in India, where seaweed diversity as well as their bioprospecting aspects, are being worked out continuously. Thus, the present work aimed to survey, collect, and identify the seaweeds of the rocky shores of coastal Karnataka in India.

2. Materials and Methods

2.1 Collection Sites

Marine macroalgal samples were collected in several collection trips to 71 localities which were divided into 11 stations based on talukas of 3 coastal districts, namely Uttara Kannada, Udupi, and Dakshina Kannada, which

extend 320 km along 74° 51'E, 12° 45'N and 74° 05'E, 14° 53'N in the west central part of peninsular India. Sampling sites were visited 1-2 hours before the low tide. GPS coordinates were recorded for each sampling site (Table 1). The map was constructed to locate the sites using Q-GIS software (Fig. 1). The Nature of the coasts was noted, and marine macroalgae were collected.

Table 1. GPS coordinates from coastal Karnataka

Name of the coastal districts	Name of taluks (Stations)	Location	Latitude	Longitude	Nature of coasts
Uttara Kannada	Karwar (UKKa01)	Tilamatti	14.900100	74.091598	S, wave-dominated, open coast
		Dandebag	14.877350	74.103174	RS, moderate tidal influence
		Kodibag	14.838772	74.132213	S, estuarine influence from Kali River, calmer waters
		Alligadda	14.801272	74.119114	S, moderate wave action
		Karwar port	14.800805	74.113091	RS, strong tidal influence
		Baithkol breakwater	14.804991	74.112295	R, artificial breakwater, strong wave action
		Vakkalakeri	14.735612	74.250012	RS, tidal Influence
	Ankola (UKAn02)	Belekeri	14.704587	74.262914	RS, high wave action, open beach
		Harikant keri	14.671213	74.279597	S, moderate tidal effect
		Mangateshwara	14.661016	74.277582	RS, strong tidal influence, island coast
		Keni island	14.654965	74.281100	RS, wave-dominated, open coast
		Shedikuli	14.649142	74.282117	S, moderate tidal influence
		Honnagudi	14.639005	74.284849	S, moderate wave action
		Uttara	14.626821	74.280642	S, high wave energy beach
		Kharvivada	14.617035	74.279041	S, moderate tidal effect
		Belambar	14.602730	74.290176	S, strong wave action
		Honnebail	14.542508	74.313744	RS, high wave energy, tourist beach
	Kumata (UKKu03)	Manjaguni	14.525323	74.314821	RS, enclosed bay, moderate waves
		Gokarna	14.519356	74.319496	RS, curved coastline, high waves
		Kudle	14.518255	74.337145	RS, isolated, high wave energy
		Om beach	14.520585	74.348418	R, wave-dominated
		Paradise beach	14.465673	74.373776	RS, moderate tidal effects
		Kelaginstala	14.438310	74.382941	S, high wave action
		Baada	14.430007	74.382863	S, moderate tidal effects
		Kadle	14.418462	74.386748	S, moderate tidal effects
		Vannalli	14.370997	74.404664	S, open coast, strong wave action
		Kumata beach	14.359017	74.405915	S, moderate estuarine influence, tidal influence
		Math	14.280504	74.427640	S, estuarine influence from Sharavathi River
		Ramanagindi	14.261033	74.432064	S, high wave energy beach
	Honnavar (UKHo04)	Honnavar port	14.226839	74.442809	S, moderate wave action
		Kasarkod	14.214160	74.449650	S, moderate tidal effect
		Kelaginur	14.174081	74.472457	S, high wave energy beach
		Mugali	14.140467	74.478148	S, moderate tidal influence
	Bhatkal (UKBh05)	Bedrakeri	14.094809	74.484848	RS, strong wave action, tidal influence
		Murudeshwar	14.014581	74.505941	S, estuarine influence from river
		Tenginagundi	13.998518	74.511767	S, wave-dominated coast
		Jalikodi	13.977069	74.524318	RS, moderate tidal effects
		Karikal	13.967909	74.532141	RS, high wave energy
		Bhatkal light house	13.960176	74.541522	S, tidal influence
		Nastaar	13.949164	74.548845	RS, moderate wave action
		Huyilmadi, Hadin	13.920497	74.586149	S, high wave action
Udupi	Baindur (UdBa06)	Shiroor	13.872007	74.603965	S, strong tidal influence
		Someshwar, Baindur	13.851043	74.611408	S, moderate waves
		Paduvari proper			

Dakshina Kannada	Kundapura (UdKu07)	Bijooru	13.827763	74.617069	S, moderate tidal effects
		Koderi	13.795253	74.624443	S, estuarine influence from river
		Navunda	13.761298	74.632351	S, moderate tidal effects
		Kirimanjeshwar	13.777187	74.629238	RS, wave-dominated
		Maravanthe	13.708575	74.641480	S, strong wave action, highway adjacent beach
	Udupi (UdUd08)	Gujjadi	13.678690	74.647074	S, high wave energy
		Madi	13.665359	74.651541	S, moderate waves
		Kharvi keri	13.655418	74.661607	S, estuarine influence
		Beejadi	13.576813	74.677227	S, moderate tidal effects
		Padukere	13.495757	74.686835	S, high wave energy
	Kapu (UdKa09)	Thonse	13.389144	74.696138	S, moderate wave action
		Malpe	13.361843	74.697512	S, estuarine influence from Udyavara River
		St. Mary's Island	13.379562	74.672158	R, columnar basalt formations, high wave energy
		Pithrody Udyavar	13.301606	74.718859	SM, estuarine influence
		Mattu	13.266962	74.727413	S, moderate wave action
	Manglore (DKMa10)	Kapu light house beach	13.223304	74.737080	S, high wave energy
		Uchilakere	13.192072	74.744240	S, moderate wave action
		Padubidri	13.136004	74.760949	S, strong tidal influence
		Hejamadi	13.075918	74.774155	S, high wave energy beach
		Sasihitlu	13.070471	74.777063	S, estuarine influence from Nandini River
		Mukka	13.026721	74.785161	S, moderate tidal effects
		Surathakal	13.004346	74.785161	RS, moderate wave action
		Panamburu	12.937258	74.804218	S, high wave energy, industrial port nearby
		Tannirbhvi	12.891230	74.813783	S, moderate tidal influence
		Bengre	12.847626	74.826485	SM, estuarine backwaters
		Someshwar	12.794824	74.846987	RS, strong wave action
		Peribail	12.786163	74.853683	S, moderate tidal effects
		Thalapady	12.769153	74.860575	S, high wave energy

S-Sandy; R-Rocky; RS-Rocky + Sandy; and SM-Sandy + Muddy

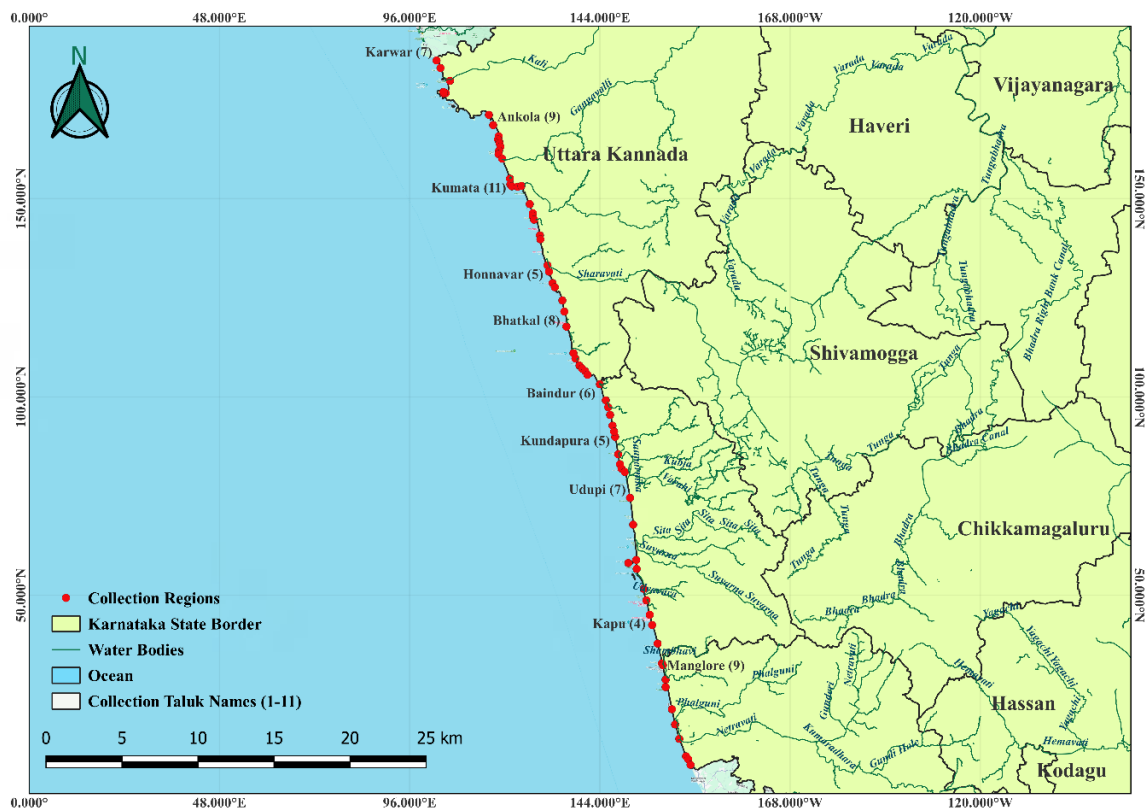


Fig. 1. Map showing collection sites in the region of coastal Karnataka

2.2 Nature of Collection Sites

Field exploration of the coastal zone of Karnataka, which is known as Kanara, a West coast of India, is sandwiched between the Western Ghats and the Arabian Sea. That comes under the Konkan coast, which includes three districts, namely Uttara Kannada, Udupi, and Dakshina Kannada, covering eleven talukas. Totally 71 sites were visited for geographical observation in the first round of field trips. It covers a wide range of geographical features, with sandy beaches and rocky beaches. These beaches or the coasts are further distinguished by their nature as given in table 1, where the nature of each location was described as sandy, rocky, sandy + rocky, and sandy + muddy. Further, the coasts can be classified generally as follows: Majali, Om beach, Vannalli coast, and most of the beaches in Uttara Kannada have headlands; pocket beaches

can be found in Tilamatti, Honnagudi, Kudle, and Mugali. Intertidal mudflats in most of the estuaries, Tudalli and Harikant Keri; Dandebag, Kodibag, Kasarkod eco beach, Baada, Manki, Kanchugodu, Kodi, Beejadi, Padukere, and Malpe are sandy beaches; sandy-gravelly beaches are in Shedikuli, Belekeri, and St. Mary's Island. Rocky islands such as St. Mary's Island have columnar basalts. Estuaries are formed by the Kali River at Karwar, the Gangavalli river at Ankola, the Sharavati river at Honnavar, and the Gurupur river at Mangalore. Mangrove belts can be found along the estuaries of Kali, Aghanashini, Mulki, Sita, Swarna, and Chakra. Cluffed coastlines like Nirvana, Nastaar, Kelaginur, and Someshwar beach Baindur; spits in Hejamadi, Paduvari proper, and Shashihittalu as well as Ottinene beach; crescents at Half Moon beach Gokarna, are prominent features (Fig. 2 & 3).

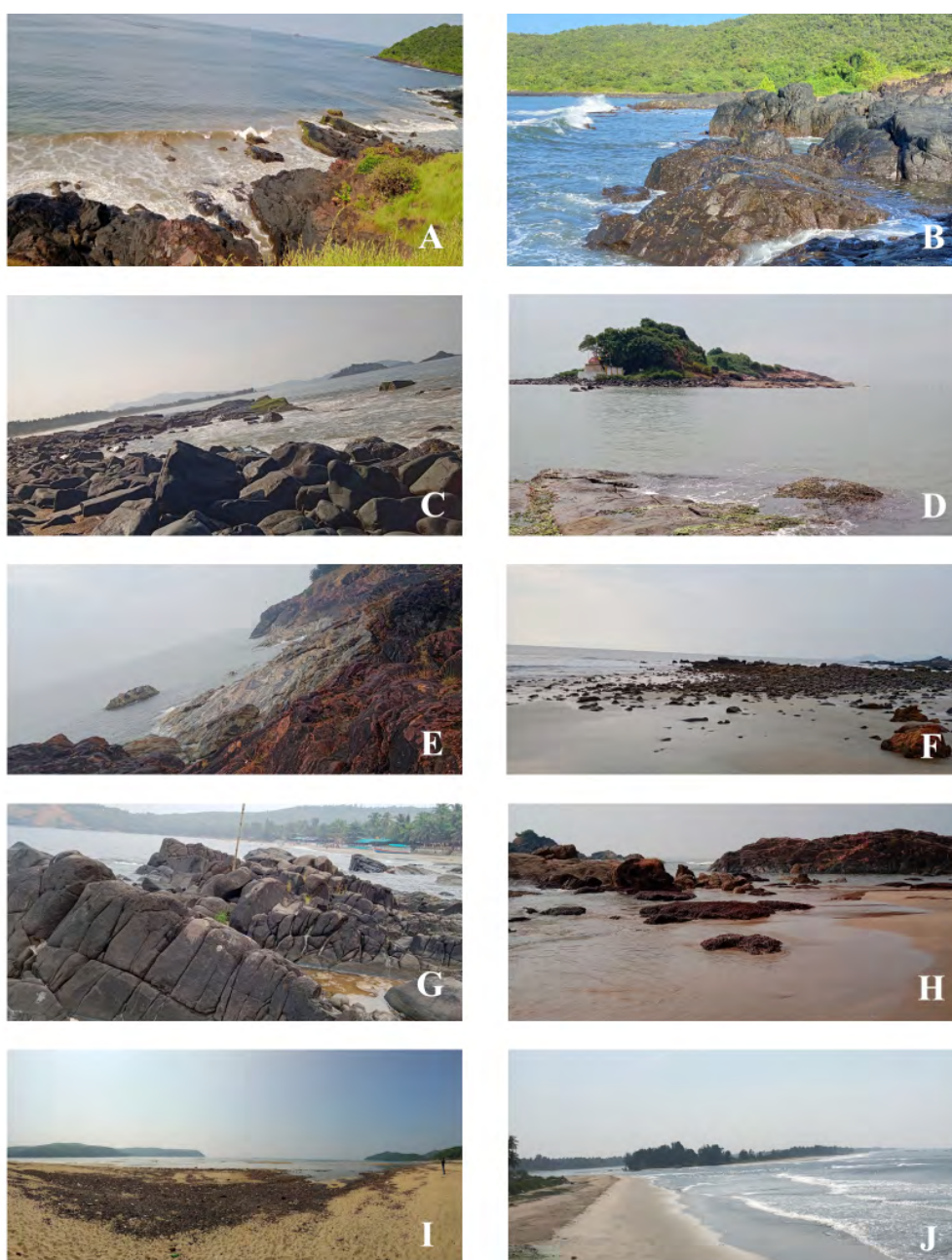


Fig. 2. Different geographical features of coastal Karnataka. (A) Majali, (B) Tilamatti, (C) Belekeri, (D) Mangateshwara Keni, (E) Vannalli, (F) Shedikuli, (G) Kudle, (H) Om beach, (I) Kelaginstala, (J) Ottinene delt

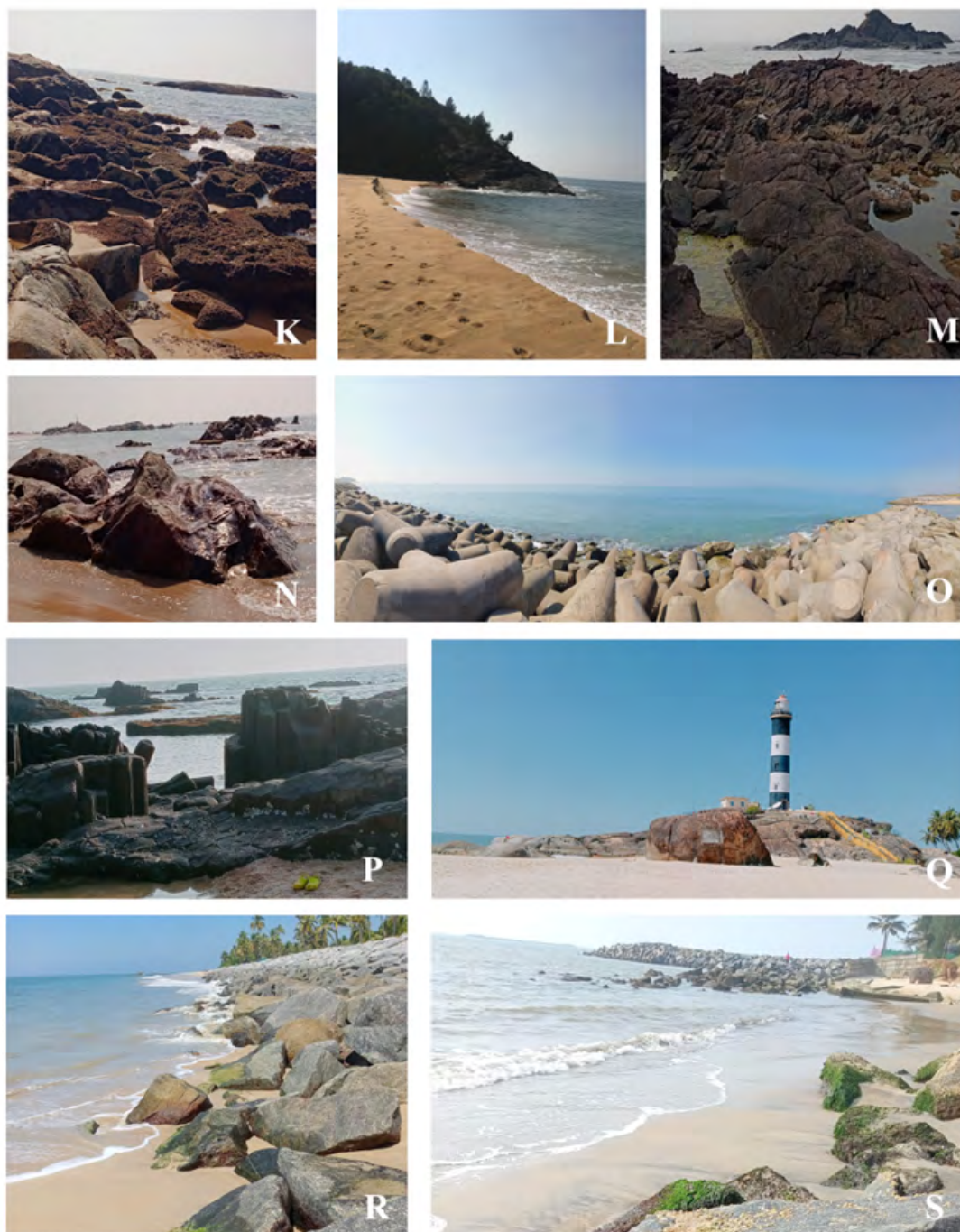


Fig. 3. Different geographical features of coastal Karnataka continued. (K) Huyilmadi, Hadin, (L) Talgod, (M) Om, (N) Gujjadi, (O) Maravanthe, (P) St. Mary's Island, (Q) Kapu light house beach, (R) Nadykere, (S) Thalapady

2.3 Collection and Identification of Marine Macroalgae

Marine macroalgae were collected from intertidal zones of the rocky shores by following a random sampling from all the possible collection modes from the substratum like natural rocks, artificial rock boulders, washouts, boats, and fishing nets, with the help of a scalpel, tweezers, scissors, and a knife. The collection trips were undertaken during November 2021 to December 2023. Initially, samples were washed in the natural habitat using seawater, then placed in plastic zip-lock bags and kept in insulated boxes until they were carried to the research laboratory as mentioned in the work of Manikandan and Vivek (2024). Collected samples

were washed 2-3 times in running tap water to remove the epiphytes and sand particles adhered to the macroalgae. Examined samples were photographed, both in their natural habitats as well as in the laboratory. Herbarium sheets were prepared for some collections for the dry mode of sample preservation for future reference according to standard protocols by Dhargalkar and Kavlekar (2004) and Botanical Survey of India (2019). Some specimens were maintained by the wet preservation method in which specimens were kept in the 5% formalin seawater solution in transparent containers. Morphological characters like the color of the thallus, length of the specimen, nature of

hold fast and habitat were the key characters observed to identify marine macroalgae. They were identified using articles, monographs, floras, and pictorial guides. Along with conventional identification methods, e-tools like Macroalgal Herbarium Portal (MHP), World Register of Marine Species (WoRMS), AlgaeBase and Global Biodiversity Information Facility (GBIF) were used for identification of collected samples (Desikachary et al., 1990, 1998; Krishnamurthy and Baluswami, 2010, 2013; Misra, 1966; Palanisamy and Yadav, 2022).

3. Results and Discussion

Marine macroalgae fall into three categories taxonomically: Chlorophytes, Phaeophytes, and Rhodophytes, generally called green, brown, and red algae, respectively, because of their pigmentation, which varies from yellow to dark brown. In the present study, a total of sixty-five species were collected and identified. Among them, twenty green algal taxa belonging to the six genera of five families (Fig. 4), namely Monostromataceae, Ulvaceae, Caulerpacae, Cladophoraceae, and Characeae, were identified. However, most of the *Chara* spp. are freshwater algae, for which they are not regarded as seaweeds. But in present study *Chara braunii* is found in the brackish water environment created on the rocks exposed to the wave actions in the rocky shore at Om coast in Gokarna.

As per traditional classification, marine macroalgae exclude microalgae and freshwater algae, but the modern molecular systematic evidence suggests that marine macroalgae are extraordinarily diverse and belong to different kingdoms, like Plantae and Chromista. Some species of *Chara*, being the lineage of land plants belonging to the Charophyceae family of Charales, are regarded as seaweed in the discussion of Bast (2014). Genus *Ulva* is diversified with ten species in the Karnataka coast, where *U. lactuca* in

Uttara Kannada and *U. fasciata* in Dakshina Kannada are abundant. *Chaetomorpha* ranks next to *Ulva* in diversity with 5 species, namely *C. antennina*, *C. aerea*, *C. crass*, *C. linum*, and *C. spiralis*, where *C. antennina* is distributed uniformly in all three districts, whereas other species are observed in particular regions as *C. spiralis* being restricted to the Udupi region. Genus *Caulerpa* thallus has creeping stolon-bearing rhizoids for attachment with erect assimilators possessing diverse shapes of ramuli based on the species (Manas et al., 2015). *Caulerpa sertularioides* with the feather-like pinnate assimilator and *Caulerpa peltata* with small peltate ramuli structures of the Caulerpacae family from the Bryopsidales order are morphologically very interesting structures among the present collections (Fig. 5; Table 2).

Fifteen genera with twenty-seven species of Rhodophyta of Florideophyceae class (Fig. 4), were identified, which are distributed within eight orders, namely Halymeniales, Rhodymeniales, Gelidiales, Gigartinales, Gracilariales, Ceramiales, Corallinales, and Sebdeniales, distributed among the thirteen families, out of which *Grateloupia* from Grateloupiaceae, *Hypnea* from Cystocloniaceae, and *Gracilaria* from Gracilariaceae are dominant and distributed evenly throughout the Karnataka coast. *Centroceras clavulatum* and *Ceramium flaccidum* are two species from the Ceramiaceae family, *Caloglossa leprieurii* of Delesseriaceae, and *Laurencia obtusa* of Rhodomelaceae from the Ceramiales order; *Amphiroa fragilissima* of Corallinales is some other collection (Fig. 6; Table 2). *Ahnfeltiopsis pygmaea* from the Phyllophoraceae family of Gigartinales order found in the Shedikuli coast of Ankola in Uttara Kannada, which is hard and rigid in appearance. *Sebdenia flabellata* from the Sebdeniaceae family had a succulent mucilaginous appearance.

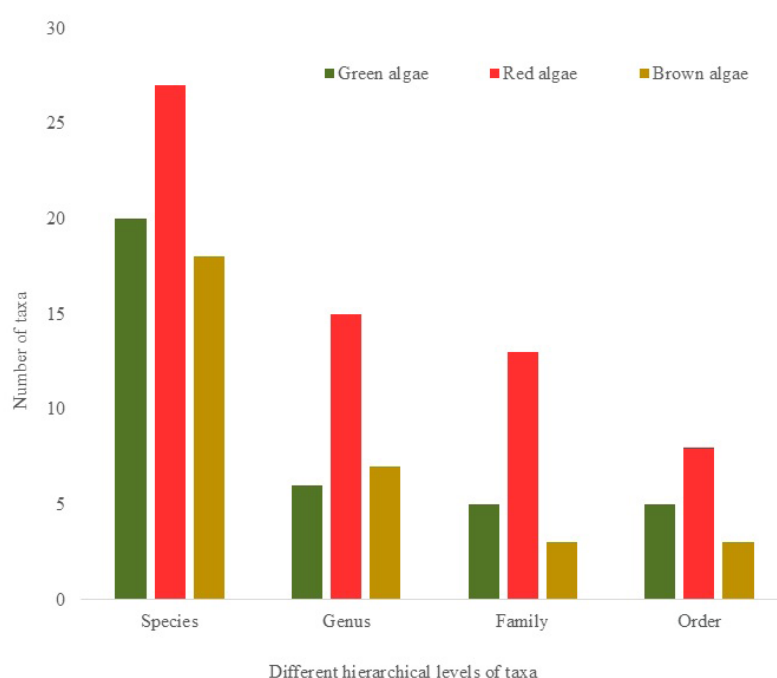


Fig. 4. Seaweed diversity among different taxonomic ranks

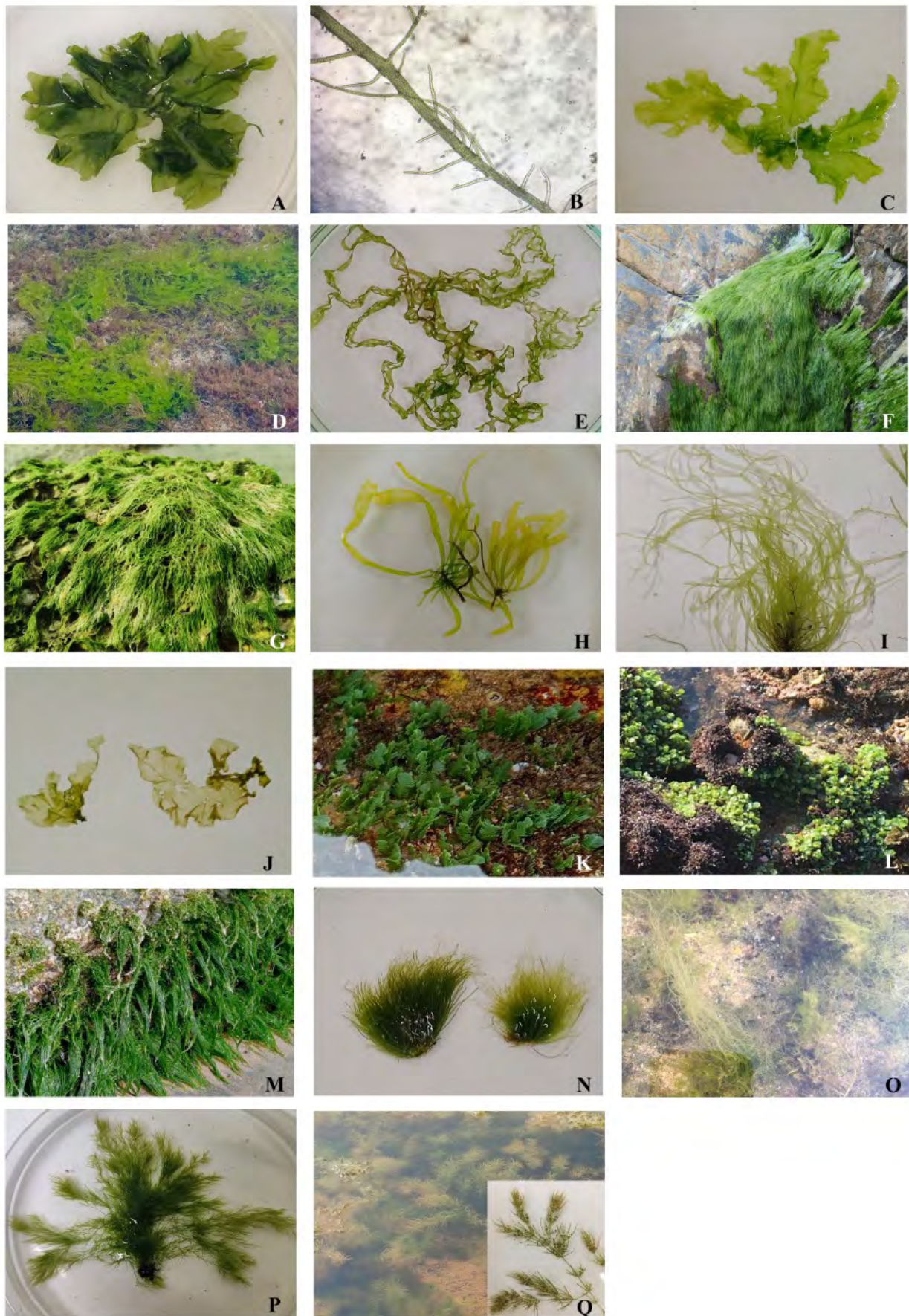


Fig. 5. Green Seaweeds of coastal Karnataka. (A) *Ulva lactuca*, (B) *Ulva clathrata*, (C) *Ulva rigida*, (D) *Ulva fasciata*, (E) *Ulva reticulata*, (F) *Ulva intestinalis*, (G) *Ulva prolifera*, (H) *Ulva compressa*, (I) *Ulva flexuosa*, (J) *Monostroma latissimum*, (K) *Caulerpa sertularioides*, (L) *Caulerpa peltata*, (M) *Chaetomorpha aerea*, (N) *Chaetomorpha antennina*, (O) *Chaetomorpha linum*, (P) *Cladophora vagabunda*, (Q) *Chara braunii*

As discussed earlier, green algae and red algae belong to the kingdom Plantae, whereas the kingdom Chromista includes brown algae in Ochrophyta as a part of the revised kingdom classification by Cavalier-Smith (1998). Eighteen taxa of Phaeophyceae members constituting three orders distributed in three families were identified (Figure 4), where *Sargassum* from the Sargassaceae family of Fucales order, with six species, is more dominant genus in brown algae. *Padina* and *Dictyota* from the Dictyotaceae family of Dictyotales order have four species in each, with even distribution among the three coastal districts, but *Spathoglossum asperum* from the same order and family is only found in Uttara Kannada. *Turbinaria conoides* exhibited a special cone-like structure differentiated among the other collections, which were the easy key identification characteristics (Fig. 7; Table 2).

As per the literature, the seaweed diversity of Karnataka was less addressed, after Kaladharan et al. (2011), who listed 78 species. Yadav and Palanisamy (2020) had updated the checklist with 105 taxa from Karnataka, of which 40% are Rhodophyceae, 32% belong to Chlorophyceae, and 28% are Phaeophyceae. The present study identified 31.25% of green, 45% of red, and 26.56% of brown algae, including the recent five new additions to the marine macroalgal flora of Karnataka reported by Chippalakatti et al. (2024). The total number of species identified was less compared to the previous report, may attributed to seasonal fluctuations, extensive drifting of seaweeds due to wave action during the senescence of seaweeds and severe coastal erosion, sedimentation in ports and harbors and municipal and industrial pollution are major concerns for declining diversity in Karnataka as well as Agadi (1985) admitted the poor vegetation of seaweeds in Karnataka is due to strong wave action along the coast.

Marine macroalgae prefer substrata to attach to for their growth; thus, they are available abundantly on the rocks as they are subjected to the harshest conditions because of desiccation, wind, wave force, and humidity. Thus, intertidal zones of the rocky shores of Karnataka are good spots for collection and diversity studies. The Uttar Kanada coast is rockier than that of Udupi and Dakshin Kannada. It is an area of complex, dynamic, and delicate environment that is also influenced by external factors like the sediment supply by the rivers and coastal processes. With varied and distinguished geographical regions and different climatic conditions, more suitable for the wide distribution of biodiversity. These observations of the geographical features of coastal Karnataka coincide with the results described by Mukhopadhyay and Karisiddaiah (2014). The tidal cycle on these coasts is semidiurnal in general, with high tide and low tide alternating twice a day with unequal amplitude, which decides the area of intertidal exposure, for which full moon days and no moon days are chosen for collection, as those have the lowest tide for the month.

Seaweeds from the late Precambrian are lithophytic, epiphytic, endophytic, and endozoic or parasitic. Most macroalgae from the present study are lithophytic, attached to rocks, shells, or pilings, such as *Chaetomorpha*, *Gracilaria*, and *Centroceras*.

Most of the *Ulva* spp. are found growing on ropes, rocks, fishing nets, and beach debris. *Chara* colonized pit-forming ponds from high tidal rocks called tidal pools in Gokarna. *Ceramium flaccidum* is epiphytic on *Gracilaria*, as most of the *Ceramium* spp. do exist. *Ulva fasciata* is found in the tidal pool only at Someshwar beach in Dakshina Kannada. Among all the sites of collection, the highest diversity is observed at Huyilmadi Hadin Eco Beach, Bhatkal in Uttara Kannada and St. Mary's Island of Malpe in Udupi. Someshwar beach at Mangalore, St. Mary's Island in Udupi, Kelaginur, Tilamatti, and most of the rocky beaches at unexplored points have abundant growth of specific seaweeds due to their geographical features and stability of hydrological characteristics and rocky structure, which support the enormous growth of seaweeds. Whereas other areas with more human activities and tourism might have disturbed the diversity due to turbidity of water and pollution. Especially *Sargassum*, *Gracilaria*, and *Ulva* spp. are abundant. Most of the *Ulva* spp. along with the *Caulerpa* sps. from the green algae are found to be edible as well as medicinally important (de Gaillande et al., 2017; Jatmiko et al., 2019). *Gellidium* sp. and *Gracilaria* sp. for Agar; *Grateloupia* sp. and *Hypnea* sp. for carrageenan; *Acanthophora* sp. and *Sargassum* sp. for agaroid; and *Sargassum* for algin are industrially important. *Ulva* spp., and *Centroceras* are medicinally important. *Ulva*, *Caulerpa*, *Padina*, *Dictyota* and *Sargassum* are used as manure (Waghmode and Khilare 2018; Yadav and Palanisamy 2019), which highlights the economic potential of seaweeds of Karnataka.

4. Conclusion

The present study on the nature of coasts and the diversity of seaweeds in rocky shores of Karnataka revealed that it covers a wide range of coasts featuring sandy beaches, rocky beaches, headlands, pocket beaches, intertidal mudflats, sandy-gravelly, columnar basalts, cliffed coastlines, spits, crescents, mangrove belts, and estuaries. A total of sixty-five species of seaweeds from twenty-eight genera of twenty-one families in sixteen orders were identified. Among them, *Ulva*, *Chaetomorpha*, and *Sargassum* are dominant genera. Uttara Kannada coasts along with the St. Mary's Island of Udupi, have more diversity as they are rockier than other coasts. The total number of species was less compared to the previous report, which may be attributed to seasonal fluctuations, drifting of seaweeds due to senescence and severe coastal erosion, sedimentation in ports and harbour as well as due to municipal and industrial pollution. Karnataka coast has poor vegetation of seaweeds due to strong wave action along the coast compared to other states, but the economic potential of seaweeds addresses the biodiversity concern as well as gains attention to work on.

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Table 2. Systematics of seaweeds from coastal Karnataka

Kingdom	Phylum	Class	Order	Family	Species	Collection codes
Plantae	Charophyta	Charophyceae	Charales	Characeae	1. <i>Chara braunii</i> C.C.Gmelin, 1826	GrCha01
	Chlorophyta	Ulvophyceae	Bryopsidales	Caulerpaceae	2. <i>Caulerpa peltata</i> J. V. Lamouroux, 1809	GrCau01
					3. <i>Caulerpa sertularioides</i> (S. G. Gmelin) M. Howe, 1905	GrCau02
			Cladophorales	Cladophoraceae	4. <i>Chaetomorpha aerea</i> (Dillwyn) Kützinger, 1849	GrChe01
					5. <i>Chaetomorpha antennina</i> (Bory) Kützinger, 1847	GrChe02
					6. <i>Chaetomorpha crass</i> (C. Agardh) Kützinger, 1845	GrChe03
					7. <i>Chaetomorpha linum</i> (O. F. Müller) Kützinger, 1845	GrChe04
					8. <i>Chaetomorpha spiralis</i> Okamura 1903	GrChe05
					9. <i>Cladophora vagabunda</i> (Linnaeus) Hoek, 1963	GrCla01
			Ulotrichales	Monostromataceae	10. <i>Monostroma latissimum</i> Wittrock, 1866	GrMon01
					11. <i>Ulva clathrata</i> (Roth) C. Agardh, 1811	GrUlv01
			Ulvales	Ulvaceae	12. <i>Ulva compressa</i> Linnaeus, 1753	GrUlv02
					13. <i>Ulva conglobata</i> Kjellman, 1897	GrUlv03
					14. <i>Ulva fasciata</i> Delile, 1813	GrUlv04
					15. <i>Ulva flexuosa</i> Wulfen, 1803	GrUlv05
					16. <i>Ulva intestinalis</i> Linnaeus, 1753	GrUlv06
					17. <i>Ulva lactuca</i> Linnaeus, 1753	GrUlv07
					18. <i>Ulva prolifera</i> O. F. Müller, 1778	GrUlv08
					19. <i>Ulva reticulata</i> Forsskal, 1775	GrUlv09
					20. <i>Ulva rigida</i> C. Agardh, 1823	GrUlv10
			Ceramiales	Ceramaceae	21. <i>Centroceras clavulatum</i> (C. Agardh) Montagne, 1846	ReCen01
					22. <i>Ceramium flaccidum</i> (Kuetz.) Ardiss.	ReCer01
					23. <i>Caloglossa leprieurii</i> (Montagne) G. Martens, 1869	ReCal01
					24. <i>Acanthophora muscoides</i> (Linnaeus) Bory de Saint-Vincent, 1828	ReAca01
					25. <i>Acanthophora spicifera</i> (M. Vahl) Børgesen, 1910	ReAca02
					26. <i>Laurencia obtusa</i> (Hudson) J.V. Lamouroux, 1813	ReLau01
			Corallinales	Corallinaceae	27. <i>Amphiroa fragilissima</i> (Linnaeus) J.V. Lamouroux, 1816	ReAmp01
					28. <i>Gelidium micropterum</i> Kützinger, 1868	ReGel01
			Gelidiales	Gelidiaceae	29. <i>Gelidium pusillum</i> (Stackhouse) Le Jolis, 1863	ReGel02
					30. <i>Gelidiella acerosa</i> (Forsskal) Feldmann & Hamel, 1934	ReGed01

Chromista	Ochrophyta	Phyophyceae	Gigartinales	Cystocloniaceae	31. <i>Hypnea esperi</i> Bory de Saint-Vincent, 1828	ReHyp01
					32. <i>Hypnea musciformis</i> (Wulfen) J.V.Lamouroux, 1813	ReHyp02
					33. <i>Hypnea spinella</i> (C.Agardh) Kützinger, 1847	ReHyp03
					34. <i>Hypnea valentiae</i> (Turner) Montagne, 1841	ReHyp04
				<i>Gigartinaeae</i>	35. <i>Chondracanthus acicularis</i> (Roth) Fredericq, 1993	ReCho01
					36. <i>Ahnfeltiopsis pygmaea</i> (J.Agardh) P.C.Silva & DeCew 1992	ReAhn01
			Gracilariales	Gracilariaceae	37. <i>Gracilaria cortica</i> (J. Agardh) J. Agardh, 1852	ReGra01
					38. <i>Gracilaria cortica</i> var. <i>cylindrica</i> Umamaheswara Rao, 1974	ReGra02
					39. <i>Gracilaria foliifera</i> (Forsskål) Børgesen, 1932	ReGra03
			Halymeniales	Grateloupiaceae	40. <i>Gracilaria verrucosa</i> (Hudson) Papenfuss, 1950	ReGra04
					41. <i>Grateloupia indica</i> Børgesen, 1932	ReGrt01
					42. <i>Grateloupia lithophila</i> Børgesen, 1938	ReGrt02
					43. <i>Grateloupia filicina</i> var. <i>bipinnata</i> P.Crouan & H.Crouan, 1878	ReGrt03
			Rhodymeniales	Lomentariaceae	44. <i>Grateloupia prolifera</i> J.Agardh 1876	ReGrt04
					45. <i>Gelidiopsis variabilis</i> (Greville ex J.Agardh) F.Schmitz, 1895	ReGes01
			Sebdeniales	Sebdeniaceae	46. <i>Gelidiopsis repens</i> (Kützinger) Weber Bosse, 1928	ReGes02
					47. <i>Sebdenia flabellata</i> (J.Agardh) P.G.Parkinson, 1980	ReSeb01
	Dictyotales	Dictyotaceae	Dictyotales		48. <i>Dictyota cervicornis</i> f. <i>pseudohamata</i> (A.B.Cribb) I.Hörnig, R.Schneiter & Prud'homme van Reine, 1992	BrDic01
					49. <i>Dictyota ceylanica</i> Kützinger, 1859	BrDic02
					50. <i>Dictyota ciliolata</i> Sonder ex Kützinger, 1859	BrDic03
					51. <i>Dictyota dichotoma</i> (Hudson) J.V.Lamouroux, 1809	BrDic04
					52. <i>Padina boergesenii</i> Allender & Kraft, 1983	BrPad01
					53. <i>Padina boryana</i> Thivy, 1966	BrPad02
					54. <i>Padina pavonica</i> (Linnaeus) Thivy, 1960	BrPad03
					55. <i>Padina tetrastromatica</i> Hauck, 1887	BrPad04
					56. <i>Spathoglossum asperum</i> J.Agardh, 1894	BrSpa01
					57. <i>Stoechospermum marginatum</i> (C.Agardh) Kützinger, 1843	BrSto01
			Fucales	Sargassaceae	58. <i>Sargassum cinctum</i> C. Agardh, 1848	BrSar01
					59. <i>Sargassum cinereum</i> J.Agardh, 1848	BrSar02
					60. <i>Sargassum cristaeifolium</i> C. Agardh	BrSar03
					61. <i>Sargassum polycystum</i> C.Agardh, 1824	BrSar04
					62. <i>Sargassum tenerimum</i> C.Agardh, 1848	BrSar05
					63. <i>Sargassum wightii</i> Greville, 1848	BrSar06
					64. <i>Turbinaria conoides</i> (J.Agardh) Kützinger, 1860	BrTur01
			Sphacelariales	Sphacelariaceae	65. <i>Sphacelaria rigida</i> Kützinger, 1843	BrSph01

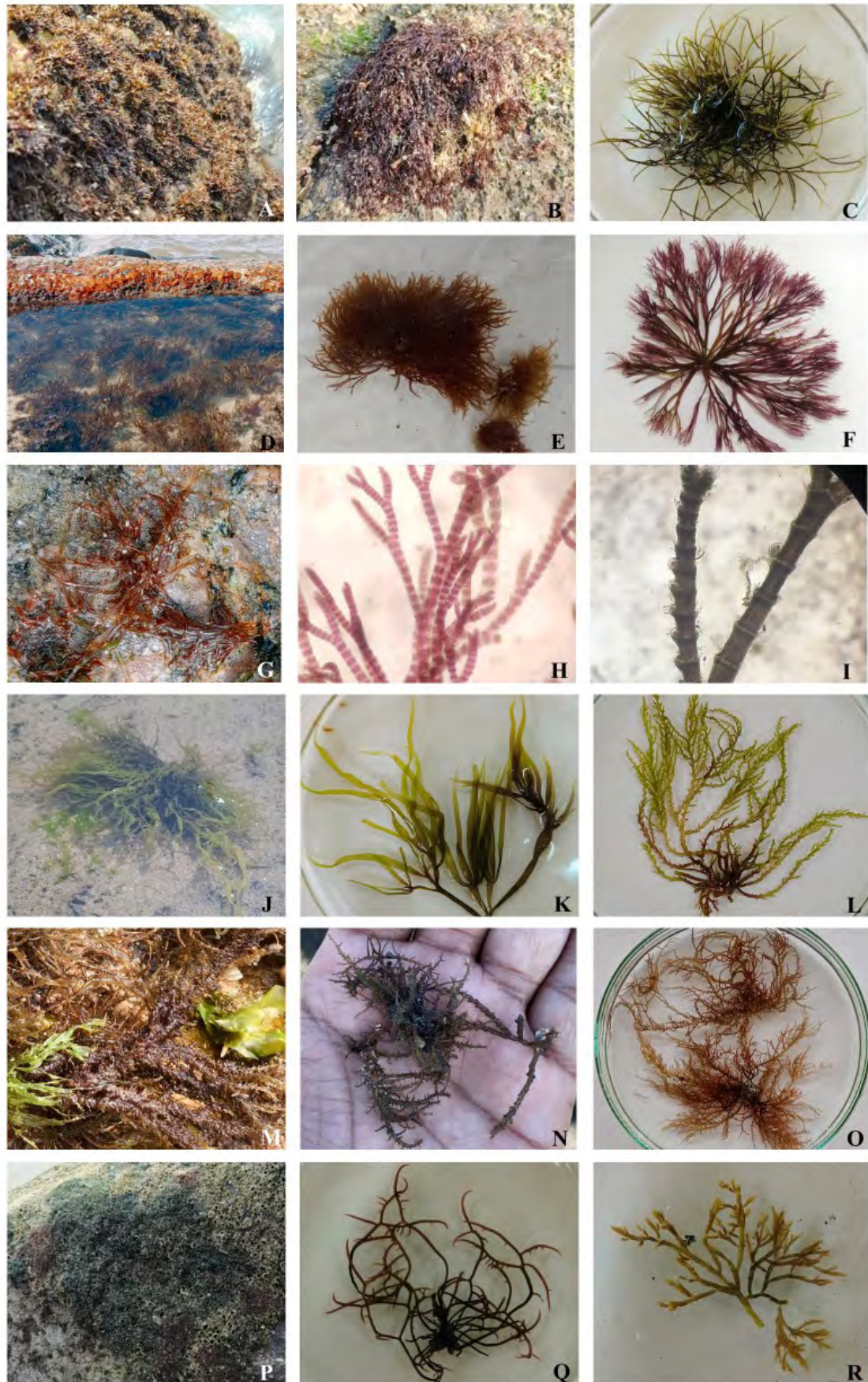


Fig. 6. Red seaweeds of coastal Karnataka. (A) *Gelidium pusillum*, (B) *Gelidium micropterum*, (C) *Gelidiopsis repens*, (D) *Gracilaria foliifera* (E) *Gracilaria cortica* var *cylindrica*, (F) *Gracilaria verrucosa*, (H) *Ceramium flacidatum*, (I) *Centroceras clavulatum*, (J) *Grateloupia filicina*, (K) *Grateloupia lithophila*, (L) *Acanthophora spicifera*, (M) *Acanthophora muscoides*, (N) *Hypnea spinella* (O) *Hypnea esperi*, (P) *Caloglossa leprieurii*, (Q) *Chondracanthus acicularis*, (R) *Laurencia obtuse*



Fig. 7. Brown seaweeds of coastal Karnataka. (A) *Padina tetrastrum*, (B) *Padina boergesii*, (C) *Padina pavonica*, (D) *Padina borayana*, (E) *Stoechospermum marginatum*, (F) *Dictyota dichotoma*, (G) *Dictyota ciliolata*, (H) *Dictyota cervicornis*, (I) *Dictyota celynica* (J) *Sargassum tenerrimum*, (K) *Sargassum cinctum*, (L) *Sargassum cristaeifolium*, (M) *Sargassum cenerium*, (N) *Spatoglossum asperum*, (O) *Sphacelaria rigidula*

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